

Field of study: Electronics
Speciality: Advanced Applied Electronics

Master Thesis

Title of Thesis:
**Design and analysis of free-space optical data
link using SFP modules**

AUTHOR:

Kacper Ugorny

SUPERVISOR:

dr inż. Aleksander Głuszek

Streszczenie

Niniejsza praca opisuje architekturę i realizację autorskiego optycznego aparatu nadawczo-odbiorczego, wykonanego z przystępnie cenowych szeroko dostępnych elementów. Przy użyciu dwóch takich apartów możliwe jest stworzenie wysoko przepustowego łącza optycznego z aktywną kompensacją, co zostało opisane w pracy. Opisane zostały również badania takowego łącza wraz z wnioskami i możliwym kierunkiem rozwoju pracy.

Abstract

This paper describes the architecture and implementation of a custom-designed optical transceiver, constructed from affordable, widely available components. Using two such transceivers, it is possible to establish a high-throughput optical link with active compensation, what have been described in the thesis. The testing of this link is also described, along with the conclusions and possible directions for future work.

Acknowledgements go here.

Contents

1. List of Acronyms	7
2. Introduction	8
2.1 Background	8
2.2 Motivation	8
2.3 Thesis Objectives	9
2.4 Scope and Structure	9
3. Theoretical Background and State of the Art	10
3.1 Principles of FSO Communication	10
3.2 Light Propagation and Environmental Factors	11
3.3 SFP Modules and Digital Diagnostic Monitoring (DDM)	11
3.4 Existing Beam Steering Techniques	12
4. System Architecture and Hardware Design	13
4.1 Transceiver and Link Overview	13
4.2 Mechanical and Optical Design	14
4.3 Controller Design	18
5. Active Alignment and Control Algorithm	21
5.1 DDM Data Extraction	21
5.2 Connection Seeking Algorithm (Spiral Search)	22
5.3 Optimizing Algorithm	23
5.4 Control Algorithm Design	24
5.5 Misalignment Compensation	25
6. Experimental Methodology	27
6.1 Experimental Setup	27
6.2 Test Scenarios and Measurement metrics	28
7. Results and Analysis	29
7.1 Baseline Performance	29
7.2 Active Compensation Algorithm Performance	30
7.3 Environmental Impact	30
7.4 Discussion	32
8. Conclusion and Future Work	33
8.1 Summary of Achievements	33

8.2	Future Work	33
	Bibliography	35
	A. Appendices	36
	Appendix A: Bill of Materials	36
	Appendix B: Source Code/CAD raport	36

1. List of Acronyms

FSO	Free space optics
RF	Radio frequency
ISM	Industrial, Scientific, Medical Band - Free to use
DDM	Digital Diagnostic Monitoring
OOK	On Off Keying
NRZ	Non-Return-To-Zero
RZ	Return-to-Zero
LOS	Line Of Sight
SFP	Small Form-factor Pluggable
EMA	Exponential Moving Average

2. Introduction

In a time of rapid global digitalization and great increase of need to process data, demand for high throughput and reliable communication systems keeps growing. With the expand of data rates, the need for security also rises. Classic Radio frequency (RF) systems, although widely used, might not satisfy needs of all customers, due to high traffic on unlicensed Industrial, Scientific, Medical Band - Free to use (ISM) bands and possible unauthorized listening.

On the other hand there might be a situation where fiber might be also too expensive or just too difficult to install.

Free space optics (FSO) communication might meet the requirements that RF lacks and collect the most demanding users. Using precisely focused light beam, this solution not only exceeds RF solutions in terms of throughput, but also provides great security, by high directivity and difficulty of hijacking the signal.

Those characteristics makes the FSO systems highly competitive answer for need of private high throughput connections [9].

2.1 Background

Fundamental requirement of well functioning free space optical link is maintaining uninterrupted direct line of sight, between transmitter and receiver. RF systems on the other hand diverge and bounce off the obstacles or even pass through them and still can reach the destination. Ideally FSO systems, keep the transmitted beam collimated, what allows to achieve high power density and security of transmission at the expense of need for keeping precise alignment. In practical deployments it is a challenging task because of environmental factors, such like vibrations, thermal expansion or atmospheric conditions. That's why modern FSO systems are required to run advanced tracking and compensating systems.

2.2 Motivation

Although FSO systems are well described in scientific papers, developing such device is quite complex engineering task. Main motivation of taking this topic is a need of creation affordable and functional proof of concept transceiver and establish a high throughput link. Completing such connection is challenging, because it requires knowledge of multiple

fields: designing optical system, precise mechanics, electronics and low level embedded system programming with real time requirements.

2.3 Thesis Objectives

Goal of this thesis is to design and develop a stable, affordable, high-speed FSO Ethernet link with active misalignment compensation. This will be achieved by using low-cost, readily available SFP modules and a custom-built mechanical system for beam steering, controlled by a microcontroller that reads the Digital Diagnostic Monitoring (DDM) data from the SFP module to maintain optimal alignment. Realization of this link will be used as a platform to analyze practical issues induced by environmental factors.

2.4 Scope and Structure

The rest of the document will provide information and knowledge about the theoretical background of FSO communication, the design and architecture of the system, the control algorithm used for active alignment, the experimental methodology, and finally the results and analysis of the project. At the end conclusions will be drawn, it will be stated what wasn't optimal in taken solution and what should be done to improve a solution.

3. Theoretical Background and State of the Art

As it was stated in introduction, the task of creating a transceiver and then establishing the link is quite challenging, because of involvement of multiple scientific fields, such as: precise mechanics, optics systems, electronics, motor control and low level firmware developing. This chapter will showcase how such systems are working to give a reader brief understanding of issues that need to be overcome later in the development.

3.1 Principles of FSO Communication

Generally, an FSO link is a set of two components: a transmitter and a receiver. The transmitter sends data signal in free space using modulated optical radiation. It consists of radiating element, which is usually a laser diode, a laser modulator and set of optics responsible for creating a beam.

Modulation is an essential functionality in an optical transmitter. It allows for embedding data into light beam, which then travels the significant distance to the receiver. Easiest way of modulating an optical signal is using direct modulation - changing an injection current of a laser diode, which affects the output optical power. That way on-off binary modulation can be achieved, more widely known as On Off Keying (OOK) [5]. OOK simply turns on the light source to transmit a logic "one" and turns it off to transmit a "zero". There are two types of OOK which are Non-Return-To-Zero (NRZ) and Return-to-Zero (RZ). As names suggest a RZ totally turns off the transmitter and a NRZ goes down to some level of output power. RZ has advantage of higher sensitivity [4].

Modulated light beam now enters an optical system that often consists of telescope or a collimator make the light beam capable of traveling long distances. Such signal needs to travel through an open space to the receiver. Line of sight of the transmitter and the receiver is required. There are many factors that influence the path of the light, some directly like weather conditions and other indirectly like vibrations, wind or thermal expansion. Taking those into consideration, tracking systems must be applied [2].

When the beam reaches its destination, it enters the optic systems, which focus it at the photodiode, which converts light into electrical signal. Then demodulation of signal is done and data is received on the other end of the link.

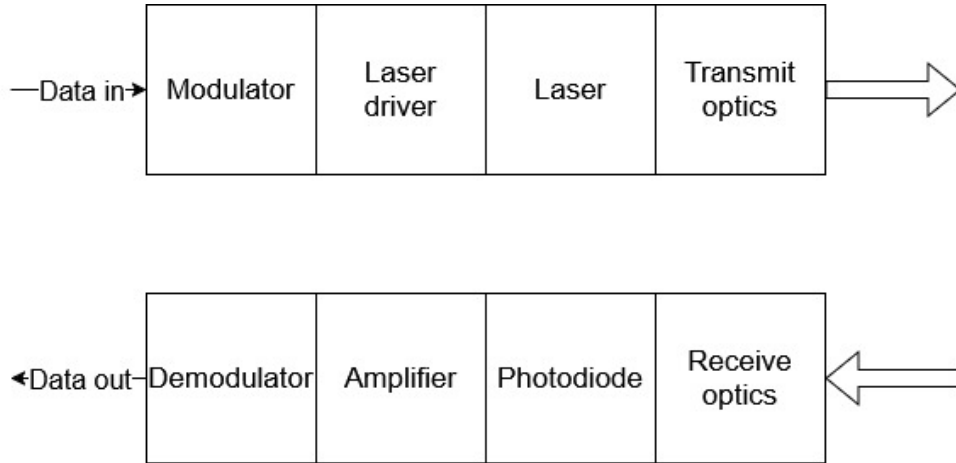


Fig. 3.1: Example FSO system

3.2 Light Propagation and Environmental Factors

Creation and development of FSO systems requires extensive knowledge about light propagation as well as awareness of environmental factors that affect the transmission. Few of the examples are described below [7]:

- Physical obstruction can temporally block the Line Of Sight (LOS). For example a bird or a leaf might fly through the path.
- Absorption is caused by the molecules of the water, that occurs in the atmosphere. Water dramatically decrease power of light passing through it. Especially fog is most limiting factor.
- Atmospheric turbulence might affect the beam, while passing through packets of air with different temperature therefore different refractive index.
- Atmospheric attenuation is a general factor which is a collection of different aspects like environmental conditions - humidity or amount of dust in the air.

3.3 SFP Modules and Digital Diagnostic Monitoring (DDM)

A Small Form-factor Pluggable (SFP) modules are standard piece of equipment that can be bought for affordable price. These are compact transceivers used in telecommunication. SFP's plugged into a switch or a media converter convert electrical ethernet signal into modulated optical signal.

It is intended to use them in fiber cable exceeding tens or even hundreds of kilometers. SFP modules have different type of connectors, like RJ45 for electrical, LC and SC for optical. As written before these modules are transceivers, allowing for full duplex

communication. It is realized by two physical optic channels or one optic channel using two different wavelengths (eg. 1550nm Rx/1310nm Tx).

Typically SFP uses wavelengths of telecommunication windows, which are 1st window - 850nm, 2nd window - 1310nm and 3rd window 1550nm. These wavelengths are great for fiber communication because of low transmission loss in fibers.

SFP modules are implementing DDM, which is management technology which lets users to monitor in real time few parameters of the transmission. Those parameters are: output optical power, received power, temperature, laser current and transceiver supply voltage.

Mentioned characteristics makes a SFP modules great candidate for base element of FSO transceiver. It drives the laser, handles the modulation and also implement the receiving part. Different connectors and wavelengths configurations allows for different designs of the transceiver. DDM allows for real time data harvest, which is essential for tracking system.

3.4 Existing Beam Steering Techniques

FSO link during its existence, might destabilize, what can lead to loss of received power, reduction of throughput and in the worst case total loss of connection. To limit the impact of environmental factors destroying the connection, providers of FSO links are implementing beam steering into their links. On top of that alignment of long range link, without any beam steering is a tedious task. At first glance it seems to be unnecessary, because in lab conditions link might work correctly, but deployment of long range link is a hard reality check.

The easiest way of providing beam steering into the design is to create a mechanical structure capable of rotating the whole transceiver. Using such mechanism can provide big coverage and it is great solution when transceivers are placed on mobile platforms for example UAV drone.

Next possible solution is implementing translation into the optical system. Movement of the lens or the mirror used to launch the beam, allows for change of angle in which the beam exits the transceiver.

Lastly the translation of optical element could be swapped with a tilt of the lens or the mirror, but tilting might introduce a increased level of difficulty, of understanding and modeling such movement.

4. System Architecture and Hardware Design

This chapter will show and explain taken engineering choices used while building a FSO link which this document is all about. It will start with general overview of the transceivers and the system in which such devices are deployable. Next the mechanical and optical design will be shown and explained. At the end the electronic circuit design will be showcased.

4.1 Transceiver and Link Overview

Transceiver architecture was carefully chosen, to be realistic to finish in given deadline and also to be competitive on the market. Another constraint was the affordability and availability of the elements on the market. Main elements which it consist of are:

- gigabit SFP media converter,
- full duplex, 1.25GB/s, 80km, WDM, SFP 1490/1550 (rx/tx or tx/rx depending on the unit) SM SC connector providing one optical path, with DDM,
- 3d mechanical movement allowing for SFP translation,
- Nema 17 1.8 deg per step
- plano convex lens 50mm diameter, 85mm focal length.
- electical control system for driving the motors and tapping into SFP's DDM,
- esp32c6, for low throughput RF transmission,

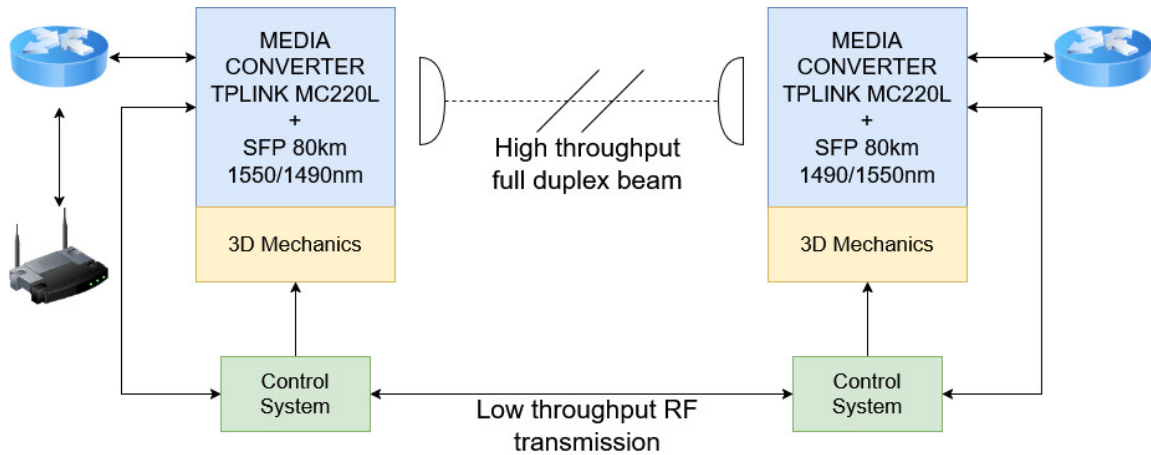


Fig. 4.1: Transceiver and System diagram

4.2 Mechanical and Optical Design

Crucial part of transceiver part is optical and mechanical design. In this particular case it is embedded together and movement provided by mechanics change the position of the light source to the lens. It changes the angle in which light exists the transceiver, allowing for beam steering as showcased on 4.2.

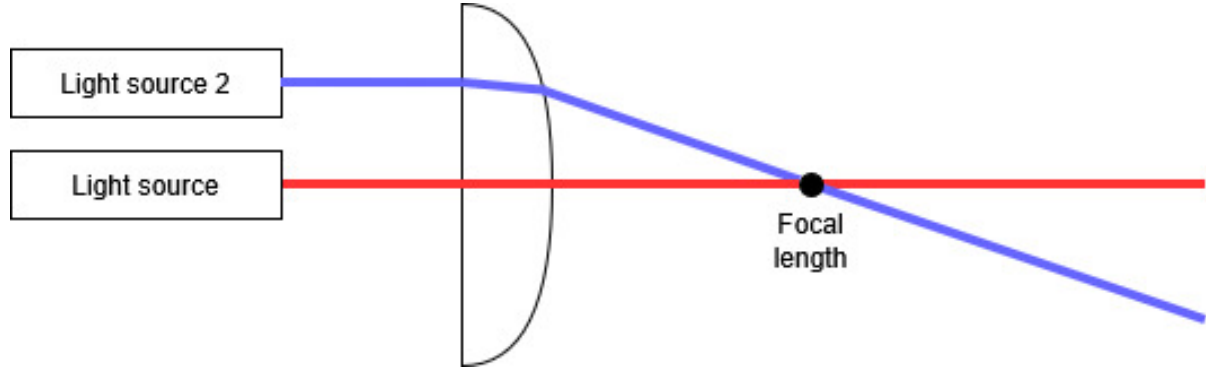


Fig. 4.2: Translation of light source against the lens

In case, when light source is placed in the focal length - the light beam will be collimated. Approximately if we move it in parallel to the lens the beam will be almost collimated and will exit the lens at the angle. If there is second light beam coming into the transceiver at the same angle as the exiting beam, it will focus in same place as the light source is. Thanks to this characteristic, this beam steering technique can work, allowing for transceivers to communicate when they are shifting from ideal alignment as shown at 4.3.

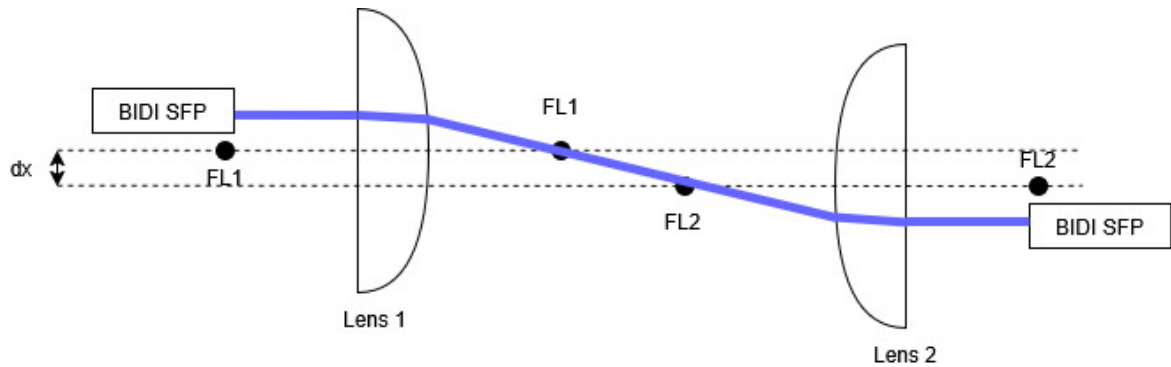


Fig. 4.3: System of 2 transceivers

To achieve such beam steering, proper mechanical system has been created, that allows for 3 axis movement of SFP module. First axis, let's call it x is perpendicular to the lens and translation alongside this axis change the distance from SFP to the lens. Another two axes, y and z are parallel to the lens and the moving the source in those direction steers the beam, as shown in the figure 4.4.

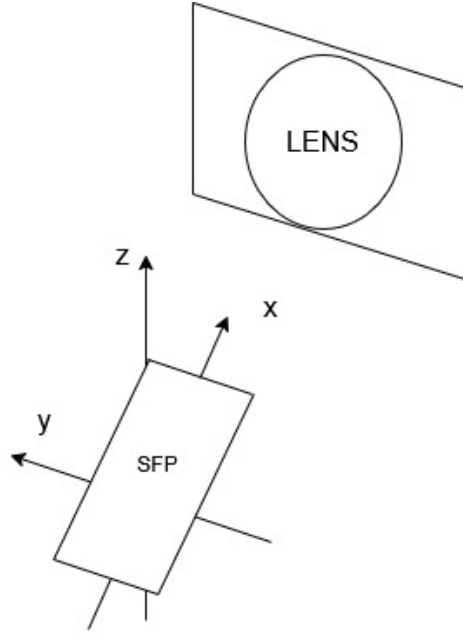


Fig. 4.4: Translations of transceiver

Mechanical structure allowing movements alongside those axes has been based on lead screws and anti-backlash nut's rotated by stepper motors. Used screw has 1 mm pitch, what means that one full rotation of the screw will move the element by 1 mm. Beside lead screw, nuts and the bearings, rest of the mechanics was design for 3D printing. While 3D printing doesn't give best properties of the mechanical structure (Low stiffness, thermal properties), it is great for rapid development and prototyping.

The 3D printed structure also includes a guides that elements move on. The shape of those guides in all axes was trapezoidal shape, as it can be seen in the figure 4.5. At the right side there is shown hole which fits the guide rails. There are circles put in the corners in order to eliminate jamming of the sharp corners.

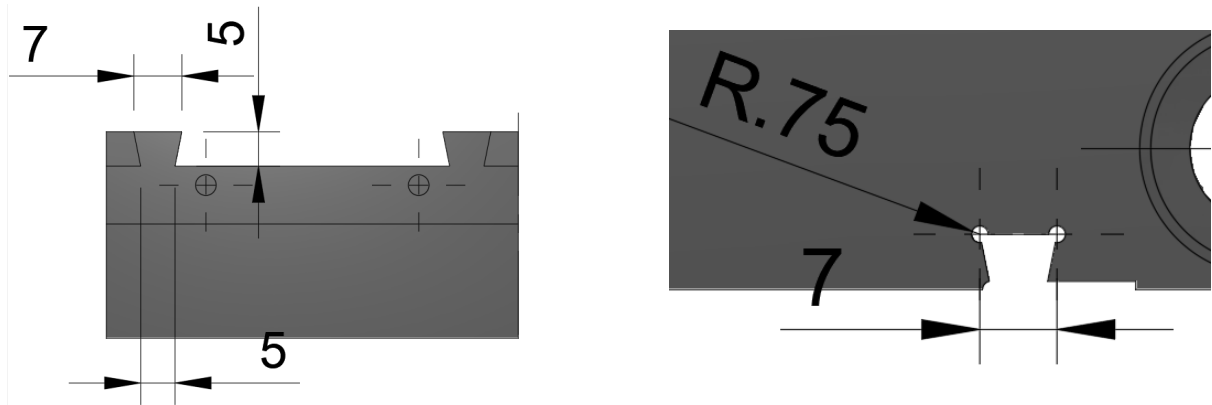


Fig. 4.5: Left: Guide dimensions. Right: Guide hole

Mechanical movement was limited to 10mm x 10mm range of movement parallel to the lens by end switches. Such range gives quite big range of motion, so it made a low range tests a bit easier. Final structure is shown in the figures 4.6, 4.7, 4.8. In the figure 4.6 the front left photo of transceiver is shown. as it can be seen lens placed in a housing and it's held in place by two screws. Also it can be noticed the whole sturcture is placed on the tripod, by custom made mount.

TODO: LEPSZE ZDJECIA TODO: OD PUNKTOW POZAZNACZAC I OPISAC NA ZDJECIACH!!!

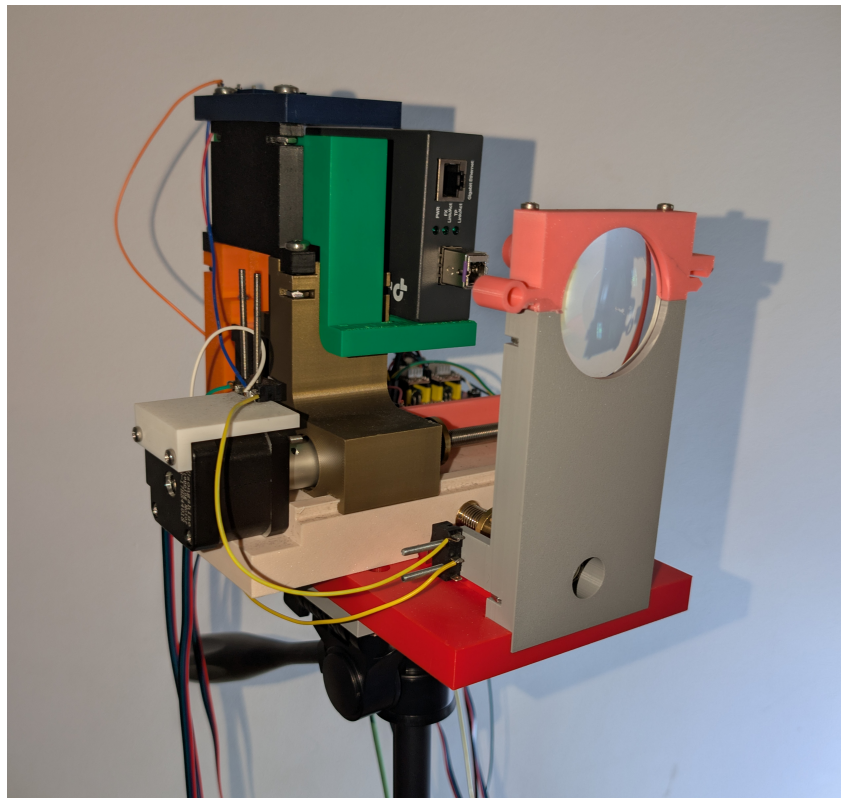


Fig. 4.6: Transceiver photo no.1

In the figure 4.7 is a picture of transceiver from other side, where lead screws are visible. There are also mechanical "Iron sights" for ease of setting up the link. On this picutre it can be seen that mechanical elements like nuts and bearings are pushed into place which fits them.

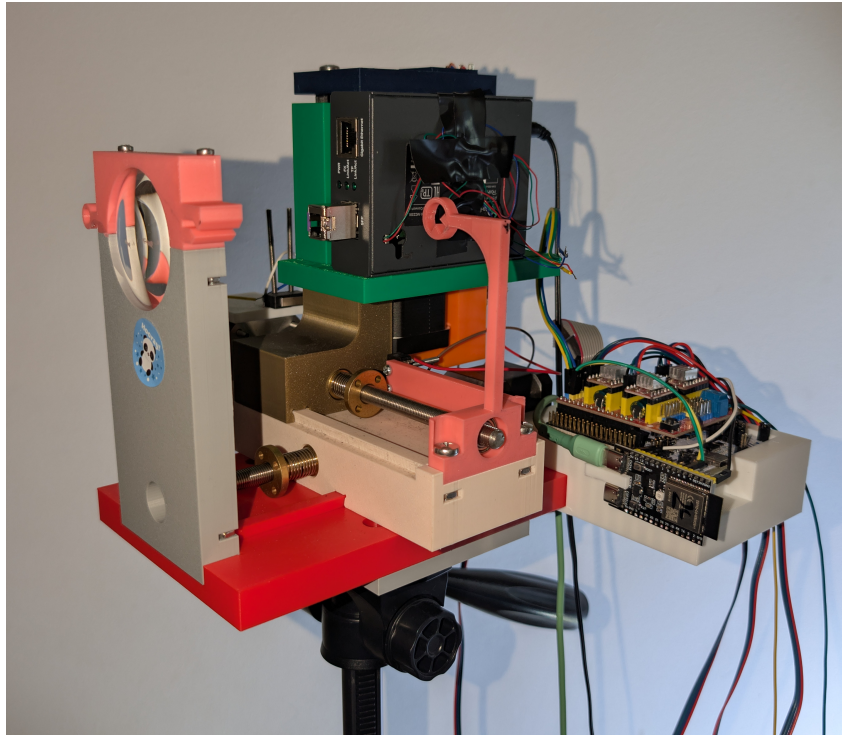


Fig. 4.7: Transcevier photo no.2

In the figure 4.8 three stepper motors are visible, these are mounted using screws into the design, also there is dedicated housing for the controller on the other end.

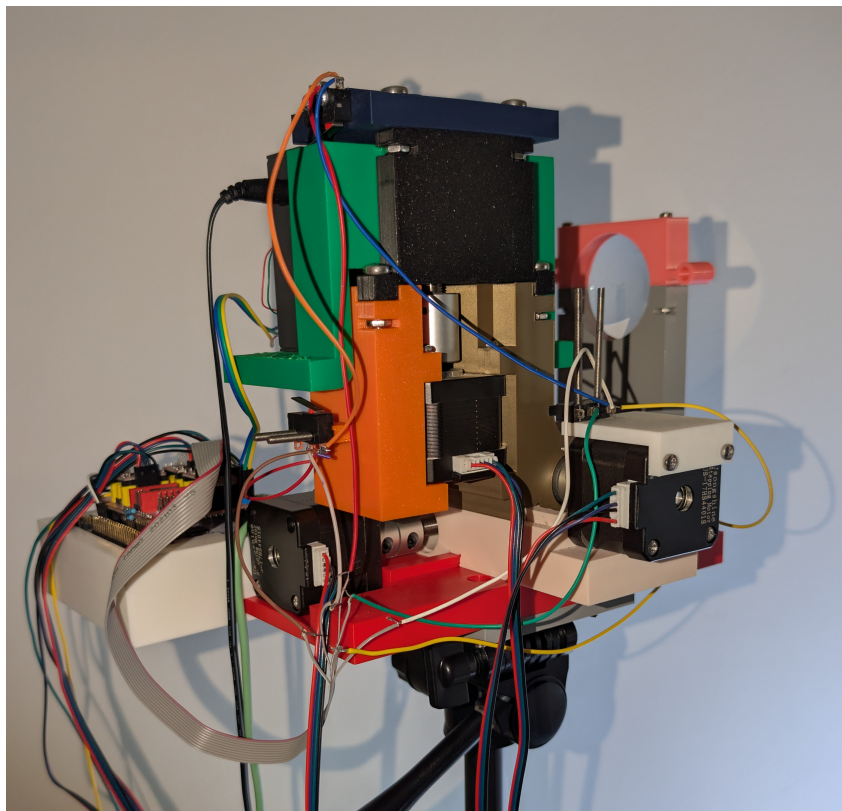


Fig. 4.8: Transcevier photo no.3

4.3 Controller Design

A controller is an important part of the project, as it's running the whole transceiver. It consists of couple elements connected by GPIO's or communication protocols.

- the heart of the controller is STM32F1 microcontroller,
- it's outputs are connected to the motor controllers A4988, which are configured for 4 x microstepping,
- inputs are used to connect the end switches connected in series, so if any of the switches of the given axes are closed then it goes to one pin,
- I2C bus is used to connect the microcontroller to the SFP module, to hijack out the DDM data,
- There is also UART connection to the ESP32 microcontroller with PCB antenna serving as RF transceiver for communication in the link.

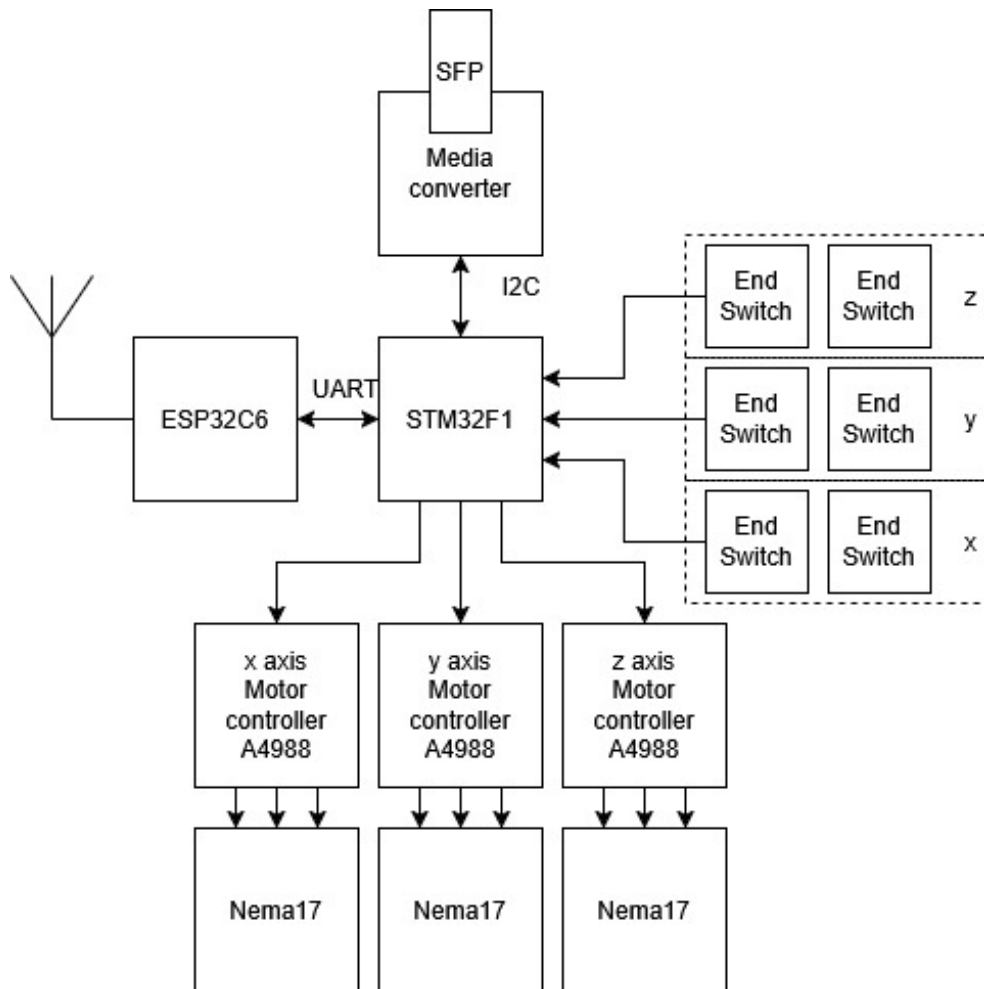


Fig. 4.9: Block scheme of the controller

The firmware, that is ran by the STM32 microcontroller, is a Non-Preemptive, Polling-Driven Single-Tasking Architecture, that calls multiple Finite State Machines (FSM). Tasks that it runs are:

- Movement Handler - it drives the motors and checks the end switches,
- Spiral Search FSM - controls the spiral search for setting up a new link,
- Homing FSM - controls the homing procedure to position the SFP in callibrated position,
- Serial Comms - checks the UART communication, from here FSM's can be toggled from sleep to run,
- DDM Readout - reads the DDM data with given period,
- Optimize FSM - algorithm that tries to optimize the received power for given distance from the lens,
- Active Compensation FSM - algorithm that keeps link at the optimal power.

It is shown graphically in figure 4.10.

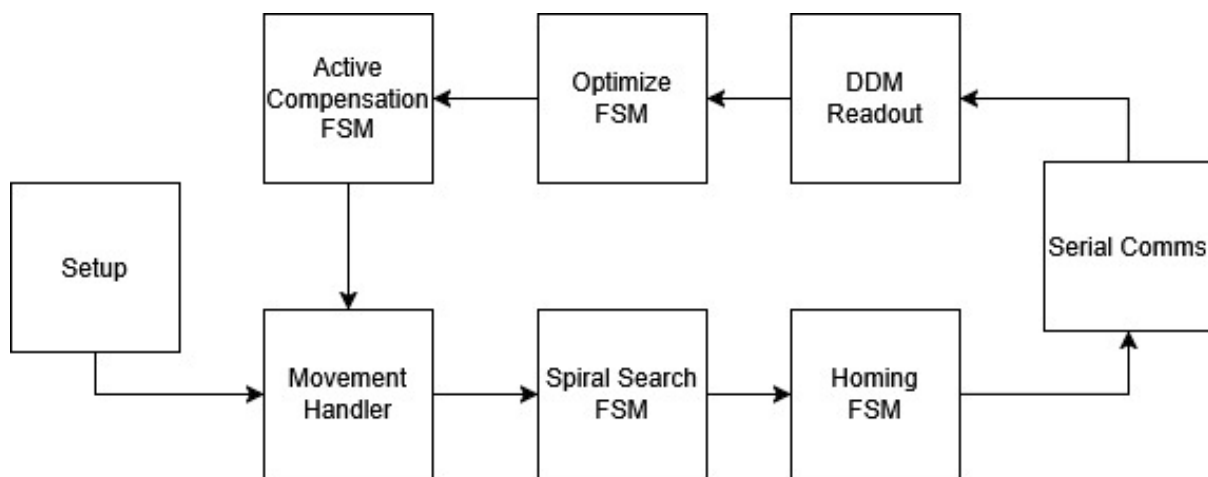


Fig. 4.10: STM32 firmware diagram

ESP32 has it own firmware, that runs freeRTOS and is responsible for information exchange with between two transceivers and lets operator connect to a link via the UART. Three concurrent tasks are ran:

- PC Serial Comms - is a task that polls for communication from PC and fowards it to the one of the transceivers,
- STM32 Serial Comms - is a task that polls for messages coming from STM32 and fowards it to the PC or second transceiver,
- ESP-NOW Message Handler - is a task that polls for messages coming from the other transceiver and redirect it to the PC or the transceiver it is connected to by a UART.

It is shown graphically in figure 4.11.

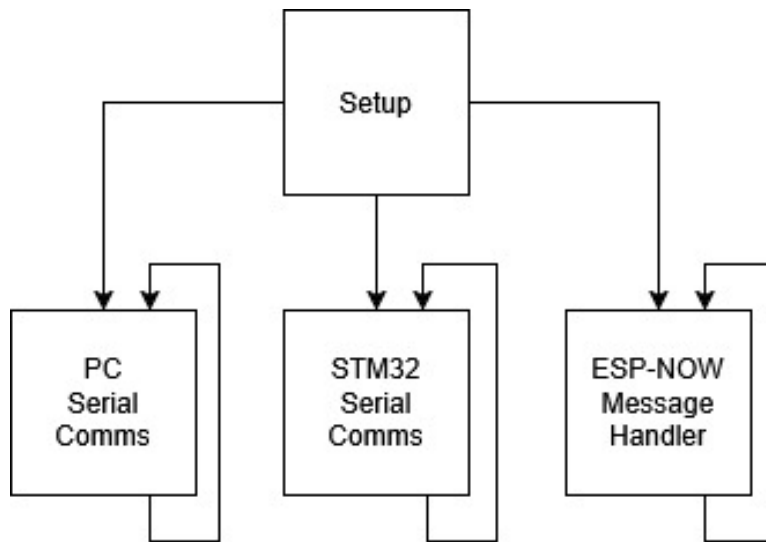


Fig. 4.11: ESP32 firmware diagram

5. Active Alignment and Control Algorithm

This chapter explains how the built platform is controlled, starting from how data is collected and then how this information is used to set up the link and keep it running.

5.1 DDM Data Extraction

As mentioned before, most of the modern SFP's has DDM implemented, which can be accessed by connecting to SFP's I2C pins, or using media converter functionalities. Unfortunately in this case, used media converter doesn't provide DDM data interface, so controller must be soldered into SFP pins.

SFP pinout is provided in the figure 5.1. SDA is at pin 6 and SCL is at pin 5.

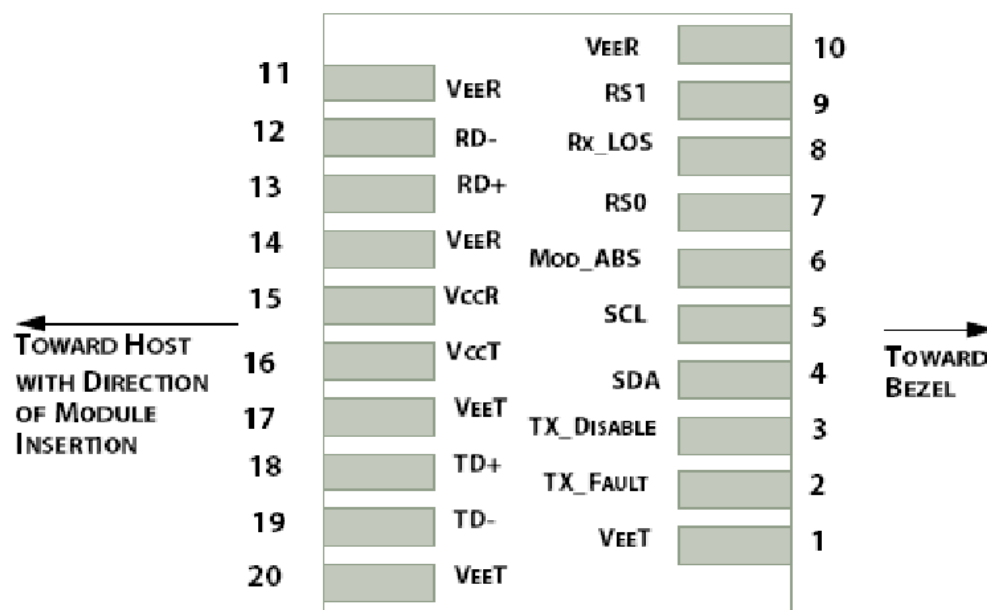


Fig. 5.1: Pinout of the sfp

There is a lot of different information about SFP module that can be accessed via this I2C, but the most interesting part is in "Real Time Diagnostic and Control Registers" part. It is set of variables that are normally used to diagnose module and link health.

Specific addresses can be seen in table 5.1.

A2h	Bit	Name	Description
Converted analog values. Calibrated 16-bit data.			
96	All	Temperature MSB	Internally measured module temperature
97	All	Temperature LSB	
98	All	Vcc MSB	Internally measured supply voltage in transceiver
99	All	Vcc LSB	
100	All	TX Bias MSB	Internally measured TX Bias Current
101	All	TX Bias LSB	
102	All	TX Power MSB	Measured TX output power
103	All	TX Power LSB	
104	All	RX Power MSB	Measured RX input power
105	All	RX Power LSB	
⋮			

Tab. 5.1: Real Time Diagnostic and Control Registers [Address A2h, Bytes 96-111] [8]

Beside this table, there is also information how to interpret those values, according to the SFF8472:

"Measured RX received optical power in mW. Value can represent either average received power or OMA depending upon how bit 3 of byte 92 (A0h) is set. Represented as a 16-bit unsigned integer with the power defined as the full 16-bit value (0-65535) with LSB equal to $0.1 \mu W$, yielding a total range of 0 to 6.5535 mW (-40 to +8.2 dBm) [8]."

There is no much information about refresh rate of this data in this specification document, so that had to be checked empirically. In case of SFP modules that were used in this transceiver, the period of updating data was $t = 15ms$, which corresponds to $f = \frac{1}{t} = \frac{1}{15 \cdot 10^{-3}} = 66.67 Hz$.

Having this information it can be concluded that with this data rate, transceiver will be capable of compensating low frequency changes.

5.2 Connection Seeking Algorithm (Spiral Search)

After placing the transceivers, so they are aimed at each other, it is though task to make them shoot perfectly "straight". It's couse imperfections of the mechanical design, or even media converter itself, the SFP module can be a little bit rotated. It can be calibrated it coarsely, but at the end some search algorithm had to be developed.

Applied algorithm is Square Spiral Search Pattern Method. Because the transceiver shoots almost straight, the direct neighborhood must be searched. Flowchart has been presented in the figure 5.2. The algorithm is straight forward, with asynchronous stop condition. Lower the dx denser the search. That dependency allows for using the search

in different scenarios, like in case the connection was lost dense version will be appropriate. During link set up rather coarse version is suggested, because the field of search is much bigger.

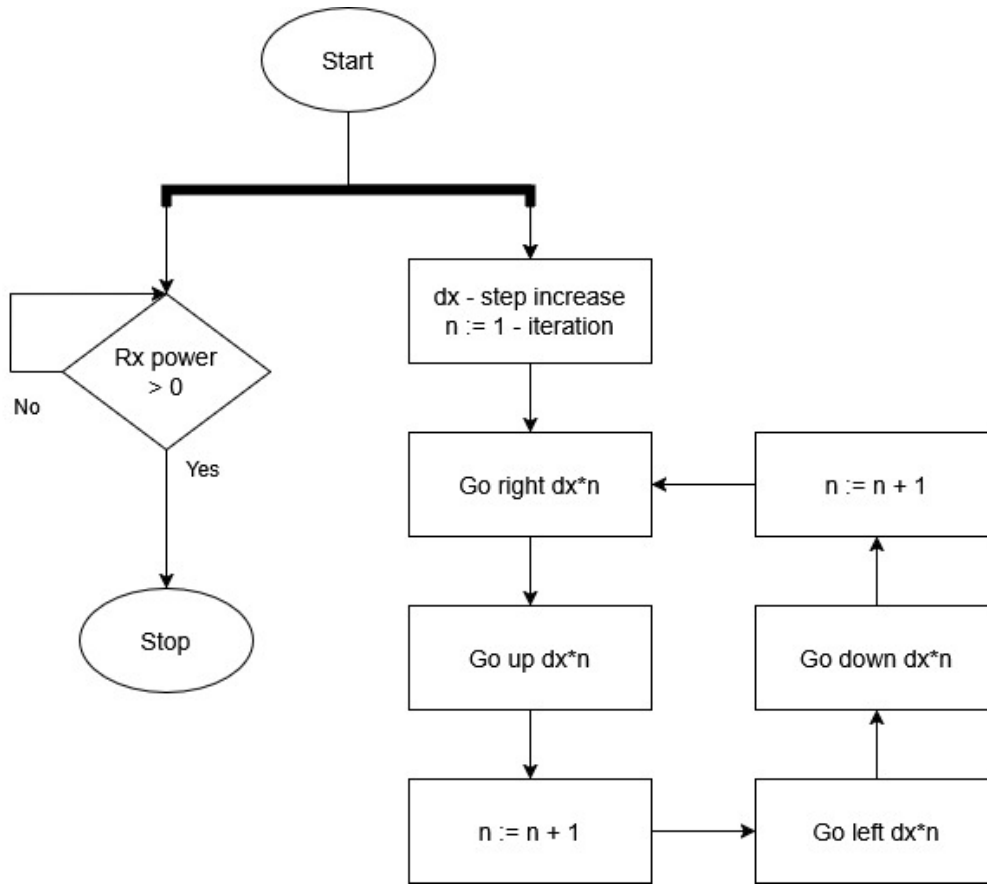


Fig. 5.2: Spiral search flow chart

5.3 Optimizing Algorithm

After running spiral search, received power is at low, unoptimal power level. The purpose of Optimizing Algorithm is to find a maximum received power point, by controlling the Y/Z axes. In this moment the continuous connection isn't required, since it's still ran during link set up.

Proposed algorithm is bounded Linear search, pseudo code is shown at listing 5.1. It sweeps three times to ensure it approach the hill from both sides and in case it starts going in wrong direction. The change direction condition is limited to quarter of current maxPow, to make the algorithm a bit faster. The 50ms delay between moving and currentPow readout lets the mechanical vibrations settle to make readout undistorted. Firstly such algorithm should be applied to one axis then another [1].

Listing 5.1: Bounded Linear Search

```

1  algorithm Bounded Linear Search is
2      currentPow := rxPow
3      maxPow := rxPow
4      dirChangeCount := 0
5      loop do
6          currentPow := rxPow
7          if currentPow >= maxPow then
8              if dirChangeCount >= 3 then
9                  return
10             maxPow := currentPow
11         else if currentPow < maxPow / 4 then
12             dir := !dir
13             dirChangeCount := dirChangeCount + 1
14         move further in dir
15         wait 50ms

```

5.4 Control Algorithm Design

When the link is already set up and used for data transfer, the continuous transmission is preferred, so a active compensation can't use same algorithm, which was used during link set up to maximize received power. As stated in DDM section, refresh rate of received power in DDM is rather slow, the algorithm to actively compensate the changing value doesn't need to be really fast.

On top of that the power curve is rather simple with one maximum, since it's the algorithm is applied in Y and Z axis (parallel to the lens), since X axis (perpendicular to the lens) should be already well align after link set up. The Discrete Hill Climing Algorithm is showcased in form of pseudo code 5.2. This procedure differentiates from the previous algorithm by the execution time, since it doesn't scan broad range, rather observe how values changes in real time. Such algorithm should be applied firstly to one axis and then to the another [3].

Listing 5.2: Hill Climbing algorithm

```

1  algorithm Discrete Space Hill Climbing is
2      currentPow := rxPow
3      dir := 0
4      dirChangeCount := 0
5      previousPow := 0
6      loop do
7          currentPow := rxPow
8          if currentPow < previousPow then
9              if dirChangeCount = 2 then
10                 return
11                 dir := !dir
12                 dirChangeCount := dirChangeCount + 1
13             previousPow := currentPow
14             move further in dir

```

5.5 Misalignment Compensation

There are few more nuances in the active compensation. Firstly, the reading from the sensor might fluctuate and can be affected by high frequency changes.

To get rid of unwanted readings, some sort of low pass filter must be applied. The LPF of choice in this case was Exponential Moving Average (EMA). It is an algorithm that doesn't require memory for saving the samples, in comparison to other moving averages like Simple Moving Average (SMA) or Weighted Moving Average (WMA) [6]. The equation for an EMA is:

$$y_i = \alpha * x_i + (1 - \alpha) * y_i \quad (5.1)$$

where y is the output, x is the input and alpha is a constant in range [0, 1].

Another point is that, link is established by two transceivers. Running algorithms at no communication between 2 ends would lead to strange results and decalibration. To make it convenient RF communication was introduced into the design. Basic handshake is introduced, as it can be seen on 5.3.

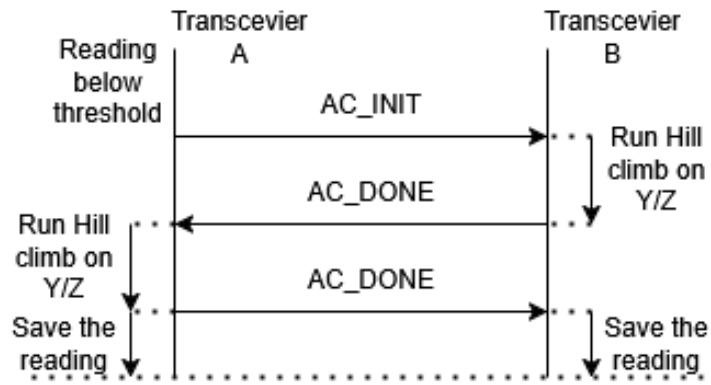


Fig. 5.3: Execution flow

Such sequence should guarantee good execution of hill climb algorithms in both of transceivers of a link. The order that transceiver initializing the compensation dispatches Hill Climing Algorithm on second transceiver firstly, ensures that no algorithm will be run when there is no working connection.

6. Experimental Methodology

In this chapter experimental methodology will be explained. The setup of the test, scenarios in which link will be placed and measurement metrics.

6.1 Experimental Setup

Transceiver itself has been equipped with custom made tripod mount, to allow for simple placement without a need for time consuming installation. The environment in which link was tested was a outdoor free-space test. Transceivers were placed in two seperate buildings and they were spaced 23 meters of each other according to rangefinder measurement, which is shown in figure 6.1.

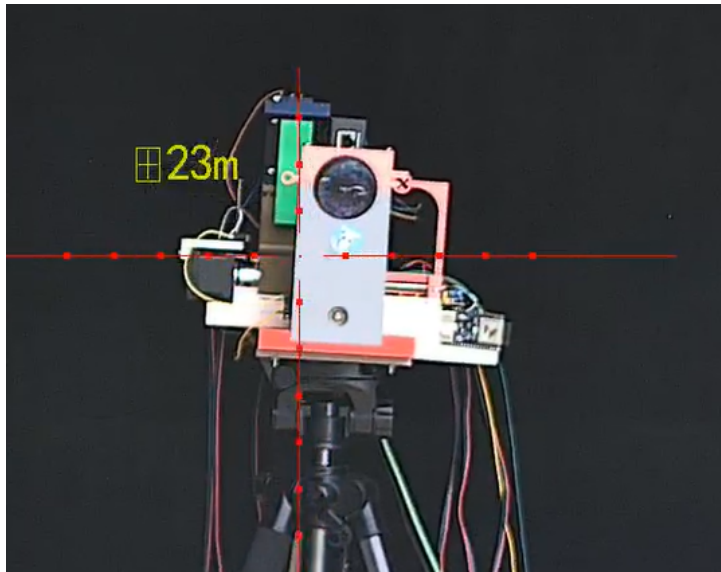


Fig. 6.1: Rangefinder measurement

Such environment is ideal for testing a link, because it's real application of such link, where the beam passes through a free space and can be affected by weather conditions. The diagram is shown in the figure 6.2. Between buildings the rain and other factors can affect the beam.

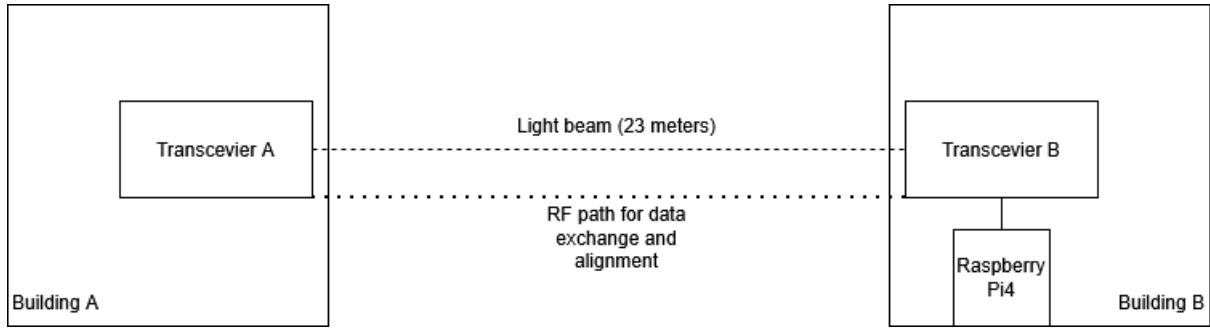


Fig. 6.2: Environment

6.2 Test Scenarios and Measurement metrics

The test scenarios were dictated by the weather conditions in this example. The conditions at the test day were variable, constantly changing, varying from clear day to runny and windy. The effects of the conditions is presented in the next chapter.

The only measurement metric, that was collected during the tests was received power. It was collected by the Raspberry Pi4 computer connected to one of the devices. The other end measurement was provided by RF interface ESPNOW.

7. Results and Analysis

7.1 Baseline Performance

The baseline for the research is performance of FSO link without active compensation turned on. Such measurements have been collected and are presented below in the figure 7.1. The blue line is a measurement from master transceiver which is the one with Raspberry PI4 connected. The orange line represent received power of second transceiver.

The results tends to slowly fall down in this case. The ripples after 800 seconds of running are caused by heavy rain.

Over the 1000 seconds of running received power decreased by almost 200uW, from 430uW in case of Master transceiver. Slave starting performance was at 200uW and didnt fall down as dramatically like master performance.

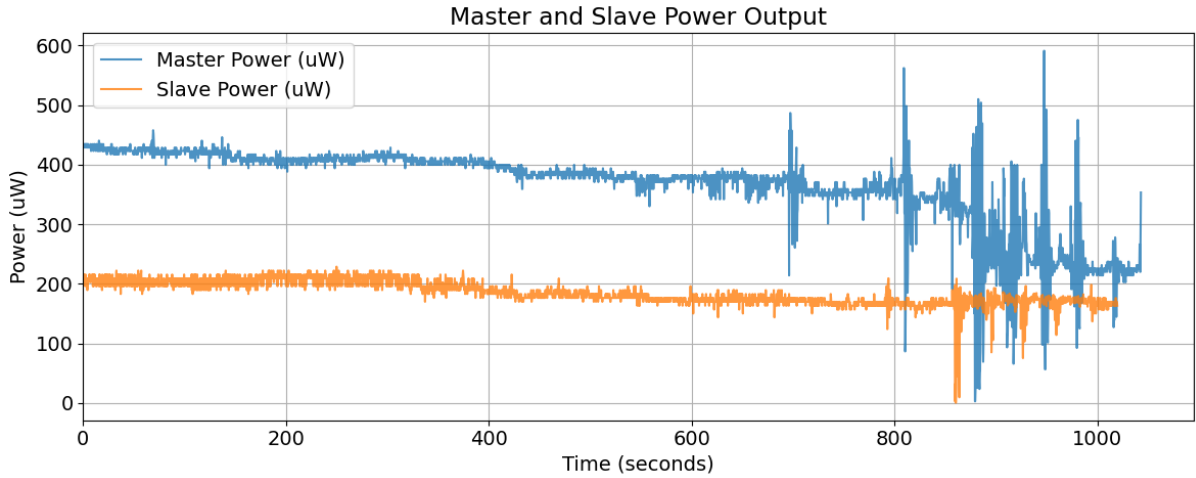


Fig. 7.1: Measurement without active compensation

7.2 Active Compensation Algorithm Performance

To compare with the baseline performance, active compensation performance was measured as well. In the figure 7.2, the performance of both Master and Slave transceivers is visible. The blue line is a measurement from Master transceiver, where the orange is the slave. On this plot there are red vertical lines that indicate AC INIT event, which is initialization of active compensation sequence.

The factors that were affecting the transceiver were weather conditions at the outside. Overallly the output power was improved throughout the measurement, by finding new maximum output power. Looking at the baseline performance, the active compensation works as intended and serves its purpose by not letting the link to degrade.

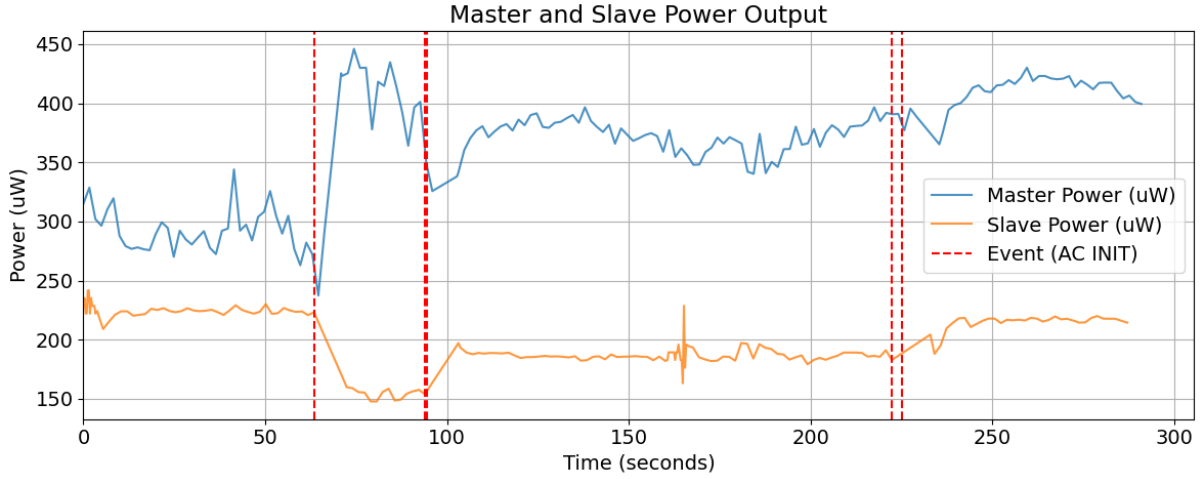


Fig. 7.2: Measurement with active compensation

7.3 Environmental Impact

As mentioned in Baseline performance case during the measurement at the end it was raining. In the figure 7.3, the last 300 seconds are presented, when the heavy rain was affecting the link quality.

Throughout the measurement there is strong variation visible at the moments of rain, even dropping the output power temporarily to 0uW. Sometimes the beam was affected in the other way, spiking the received power of Master transceiver to 600uW.

This case shows how dense amount of droplets of water might even block the connection, causing loss of packets which decrease the throughput.

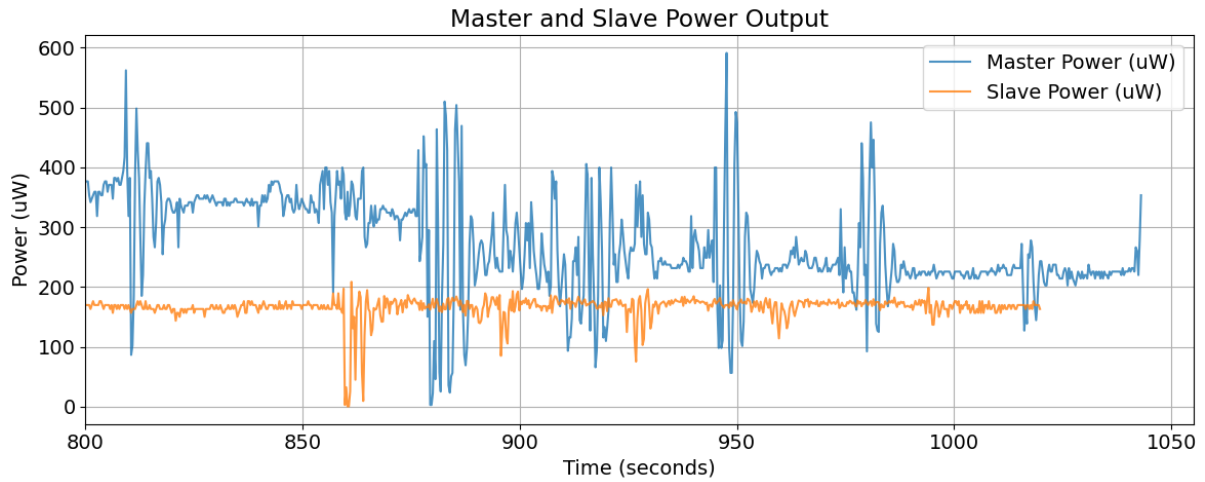


Fig. 7.3: Heavy rain effects on received power

There was another measurement in rainy day. It is 4000 second measurement starting from relatively low 50 μW received power in Master and 30 μW in Slave. The measurements in this time were effected by heavy rain with strong wind. The results shows that Master received power changed from 50 to even above 200 μW . Those results showcased moments, where received power dropped down to 0 μW and at the end the link fully destabilize.

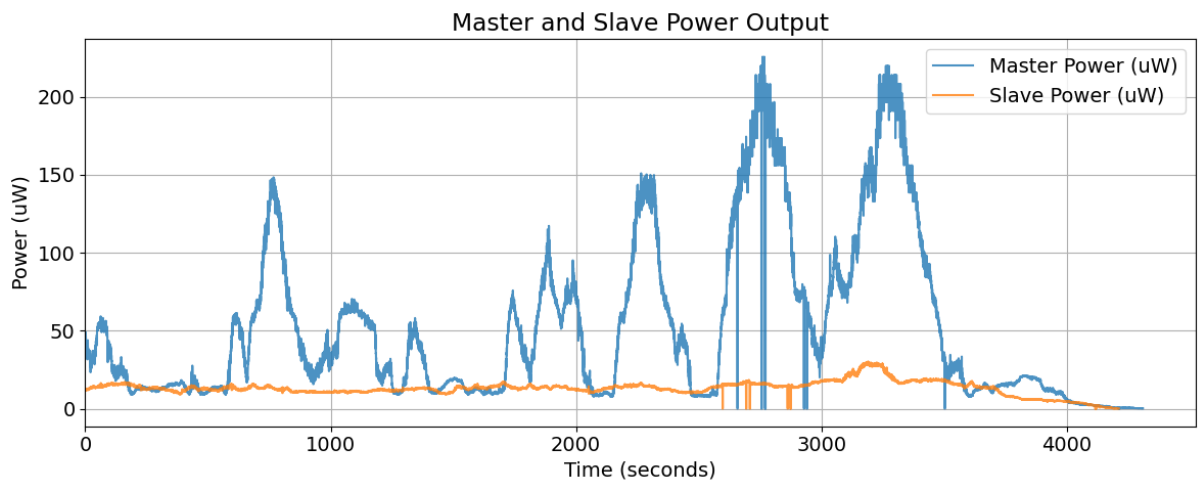


Fig. 7.4: Environment

7.4 Discussion

The results of Active Compensation in comparison to baseline performance, shows positive results - the received power metric was controlled at kept at good level. There are some nuances that can be taken in consideration while improving the overall performance. Currently the transceiver is optimizing the received power of itself, but refined version would be optimize the sum of both received powers. Biggest bottleneck of the system is the DDM polling data frequency, 15ms makes a transceivers appropriate for not moving applications. If there was an intention to apply this communication at drone to vehicle communication, the update would be too slow and would require other input data to base compensation on. Also such application would require much higher range of motion. The current mechanics could be a nice base for "fine" steering in such system were "coarse" steering would be at the base of transceiver.

Overallly the 3D printed mechanics and the steppers motors give nice stability for "last mile" applications, allowing for high throughput secure connections where it's impossible or too expensive to lay fiber. This is also a pretty good base which could be extended for more advanced application, which would run real time tracking, but that would require to swap current SFP module for custom electronics, allowing for faster data harvest.

8. Conclusion and Future Work

This chapter will conclude the work that is described in this thesis and will speculate about possible improvements of the device.

8.1 Summary of Achievements

The project was done successfully, according to the Thesis Objective. The design and produced transceiver in deed is capable of 1Gb high throughput transmission. Two of such transceivers were used to create functional FSO link. It was created using widely available components and 3D printed structure. The active misalignment compensation is shown to be working, improving the alignment in changing environment. The research was conducted, thus the impact of outdoor environment with changing weather condition was tested.

The result device came in as a complete structure, solving the issues, that were to be solved. Each part, electrical, firmware, mechanics and the optics are cooperating with each other in a intended way.

8.2 Future Work

FSO transceiver and the link is a task that can be done in various ways, according to the needed application. The transceiver described by this document aims to be applied in "Last mile" scenarios.

In such use case used components like SFP modules, are sufficient to connecting two buildings together. Although the low data refresh in DDM technology bottlenecks the possibilities of compensating different environmental factors. With current application the link is capable of reducing the slow paced changes, like thermal expansion of the buildings and the device itself. In higher distances this might start to be an issue where those fast paced vibrations or wind swings might lead to big packet losses. So the first improvement to make such transceiver greater, would be designing its own circuit of receiver and transmitter, which would allow for faster readouts of received power.

Another thing that could be upgraded are mechanics, more exactly would be changing the 3d printed guides with proper precise guide rail, which would eliminate the loose movements and decrease the friction. Current movements of the transceiver, are also far to long in tens of centimeters, limited to one centimeter range by the end switches.

Limiting it by design will reduce the amount of materials used in production, including filament and the lead screw. The production time would also be reduced by shorter 3D print times. The last thing regarding the mechanics, would be sealed waterproof casing of the transceiver, that would allow for proper application.

The last thing that would improve the device, would be beacon light source, that would allow for link set up without human interaction. Such light paired with couple photodiodes, would make the set up faster, but also would open the new possibilities of automatic link recovery in case of total link degradation.

At the end there are many little things that could be done differently or better, but those that were pointed out are the key things that would be next to improve, to make this transceiver far greater device.

Bibliography

- [1] Nitin Arora, Garima Bhasin, and Neha Sharma. Two way linear search algorithm. *International Journal of Computer Applications*, 107(21), 2014.
- [2] Scott Bloom et al. The physics of free-space optics. *AirFiber Inc. White Paper*, pages 1–22, 2002.
- [3] Sathiyaraj Chinnasamy, M Ramachandran, M Amudha, and Kurinjimalar Ramu. A review on hill climbing optimization methodology. *Recent trends in management and commerce*, 3(1):1–7, 2022.
- [4] Hennes Henniger and Otakar Wilfert. An introduction to free-space optical communications. *Radioengineering*, 19(2):203–212, 2010.
- [5] Rongqing Hui. *Introduction to fiber-optic communications*. Academic Press, 2025.
- [6] Volodymyr Lotysh, Larysa Gumeniuk, and Pavlo Humeniuk. Comparison of the effectiveness of time series analysis methods: Sma, wma, ema, ewma, and kalman filter for data analysis. *Informatyka, automatyka, pomiary w gospodarce i ochronie środowiska*, 13(3):71–74, 2023.
- [7] Aditi Malik and Preeti Singh. Free space optics: current applications and future challenges. *International journal of optics*, 2015(1):945483, 2015.
- [8] Storage Networking Industry Association (SNIA). Sff-8472 specification for management interface for sfp+. Specification Rev. 12.4, SNIA SFF Technology Affiliate Technical Work Group, 2021.
- [9] Abderrahmen Trichili, Mitchell A Cox, Boon S Ooi, and Mohamed-Slim Alouini. Roadmap to free space optics. *Journal of the Optical Society of America B*, 37(11):A184–A201, 2020.

A. Appendices

Appendix A: Bill of Materials

TODO: DODAC BOM

Appendix B: Source Code/CAD raport

TODO: dodać w wypadku generowania stron